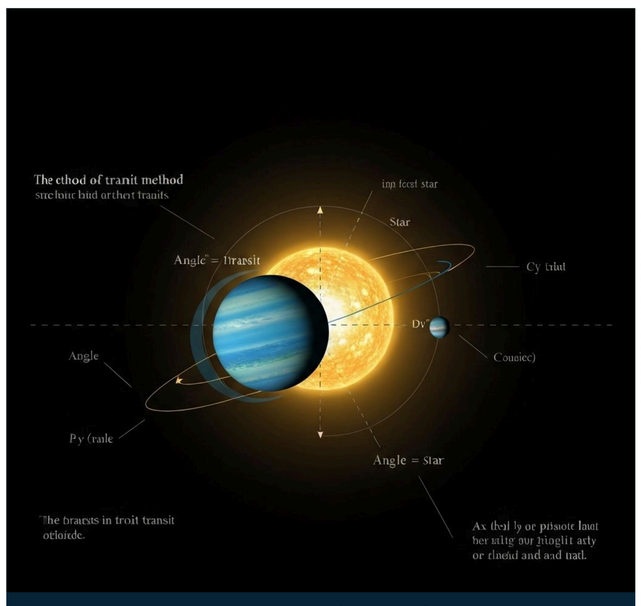
PLANET PIONEERS · AIAS+ 2026 STUDENT SHOWCASE

ExoPlain: An Interactive, Explainable AI Platform for Exoplanet Discovery

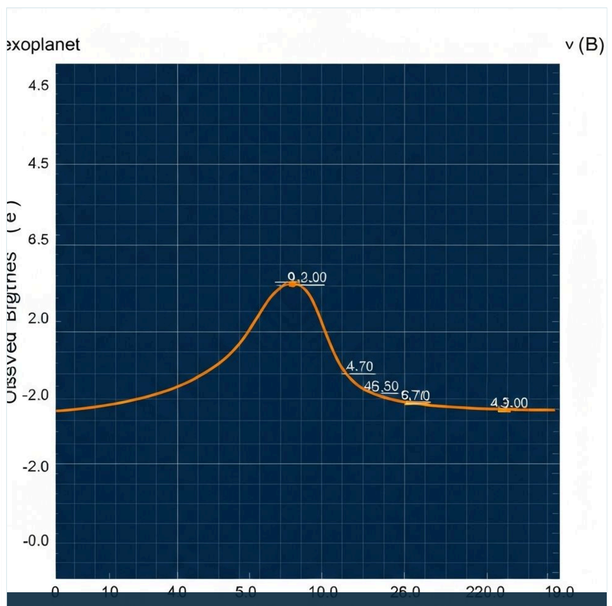
EXO PLAIN
AI FOR SCIENCE
INTERACTIVE ML

From TESS light curves and candidate parameters to interpretable evidence for human review

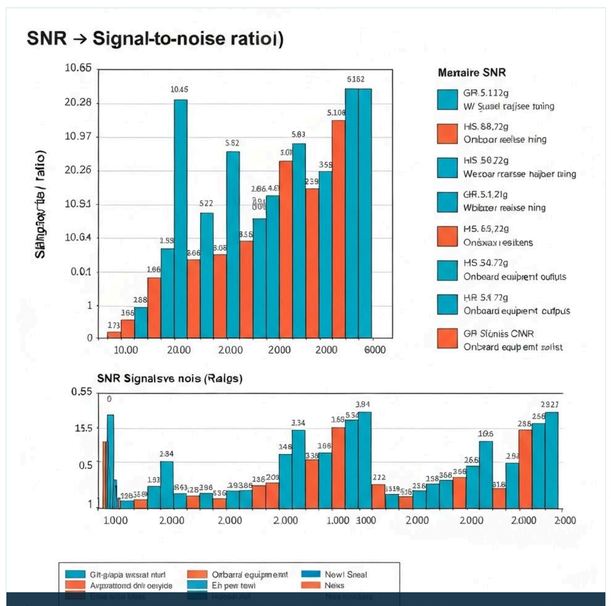
Planet Pioneers · Student research team · NASA Space Apps Challenge 2025
Authors: [student names] · School: [school] · Mentor: [mentor]



Transit geometry: a planet blocks a fraction of stellar light



Light-curve evidence: depth, duration, periodicity and noise



Quality context: weak signals require uncertainty-aware review



Interactive layer: explore examples and inspect AI explanations

01 Research problem and contribution

Transit surveys produce large time-series datasets, but a brightness dip alone does not confirm a planet. Stellar variability, eclipsing binaries, instrumental noise and processing artifacts can create similar signatures. ExoPlain combines a candidate-data interface, FITS processing, experimental classifiers and a language-model explanation layer so learners can trace how an input becomes an assessment.

Research question: How can an interactive AI system make exoplanet candidate evidence easier to interpret while preserving uncertainty and the distinction between screening and scientific confirmation?

02 End-to-end analysis pipeline

1 · Acquire
Upload a TESS/Kepler-style FITS light curve or manually enter 16 orbital, transit, stellar and sky-position fields.

2 · Clean
Select TIME and flux columns, remove NaN/Inf rows, require at least 100 valid samples and normalize flux by its median.

3 · Represent
Linearly resample to 2,048 points. Compute 37 statistical, transit, spectral, rolling, trend and autocorrelation features.

4 · Classify
Scale the feature vector and obtain Random Forest probability. CNN, LSTM and MLP scripts explore sequence-model alternatives.

5 · Explain
Convert candidate fields or labeled features into a constrained Gemini prompt and return a readable assessment with key evidence.

Two software paths currently coexist: manual parameter entry calls Gemini directly, while FITS upload calls the Python feature service and Random Forest path. The production goal is one auditable pipeline in which the classifier supplies evidence and the LLM explains it.

03 Data preparation and modeling methods

A · Labels and dataset

Training scripts use TESS light curves and TFOPWG dispositions. The code maps CP/KP/PC to the positive class and FP/Unknown to the negative class. Project documentation reports 7,638 curves with strong class imbalance, so label provenance and candidate-level grouping must be audited before publication.

```
y = 1 for CP/KP/PC; y = 0 for FP/Unknown
```

B · Signal preprocessing

The FITS service searches common time and flux columns, removes invalid samples, median-normalizes the flux and interpolates every curve to a fixed 2,048-point representation. Training augmentation adds small Gaussian noise, amplitude scaling and vertical shifts.

```
f_norm(t) = f(t) / median(f); x ∈ R^2048
```

C · Feature-based classifier

The Random Forest path standardizes 37 engineered features and uses a stratified 80/20 split with class-balanced trees. Features cover distribution shape, counts beyond 1σ-3σ, proxy transit depths, frequency-band power, trend fit and autocorrelation. The saved model stores feature importance and AUC metadata.

```
p_planet = RF(StandardScaler(ϕ(x))); threshold = 0.50
```

D · Sequence-model experiments

The 1D CNN stacks five convolution blocks (64→128→256→512→512 channels), batch normalization, pooling and dropout before a 512→256→128→64→1 head. A compact LSTM downsamples 2,048 to 256 timesteps, uses 32 hidden units and a 64→32→1 head. An MLP provides a dense baseline.

```
Loss: BCEWithLogits · Adam lr=10^-3 · weight decay=10^-4
```

E · Training controls

- Random seed 42 and 80/20 train-validation split.
- ReduceLROnPlateau halves learning rate after stalled validation loss.
- Early stopping and gradient clipping limit unstable training.
- Dropout and augmentation reduce memorization risk.

F · Transit physics represented

Candidate inputs include orbital period, epoch, impact parameter, transit duration and depth, planet radius, equilibrium temperature, insolation, SNR, stellar temperature, surface gravity, stellar radius, magnitude and sky position.

```
Transit depth δ ≈ (R_p / R★)²
```

37-feature vector: mean, standard deviation, extrema, quantiles; dip/peak counts at multiple σ thresholds; three depth proxies; variance, RMSE, skewness, kurtosis; dominant frequency/power, spectral centroid and band powers; rolling statistics; linear trend and R²; autocorrelation at lags 1/5/10 and periodicity strength.

Model	Input representation	Purpose	Main validation risk
Random Forest	37 scaled engineered features	Primary FITS-service classifier and feature-importance baseline	Feature leakage or target duplication can inflate AUC
1D CNN	Raw 2,048-point normalized sequence	Learn local transit morphology and multiscale temporal patterns	Imbalance may favor the majority positive class
LSTM	256-step average-pooled sequence	Test whether recurrent state captures longer temporal context	Downsampling may suppress short or shallow events
MLP	Flattened 2,048-point sequence	Dense neural baseline without explicit temporal structure	High parameter count and weak shift invariance

Input	Conv 7	Conv 5	Conv 3	Conv 3 × 2	Global pool + head
1 × 2048 flux	64 channels	128 channels	256 channels	512 channels	512→256→128→64→1

04 System architecture and data contracts

Next.js / TypeScript client
Manual candidate form, FITS upload, 3D galaxy exploration and results interface.

Next.js API routes
Validate requests, format candidate measurements and call the Gemini 2.5 Pro endpoint.

Python / Flask service
Parse FITS tables, normalize and resample flux, compute features, scale them and run the Random Forest.

Model artifacts
Serialized scikit-learn RF/scaler plus PyTorch CNN, LSTM and MLP checkpoints used in experiments.

INPUT	MODEL EVIDENCE	OUTPUT
FITS or 16 fields	**37 features + p**	**assessment + rationale**

05 Explainability design

Evidence selection
For manual entry, the prompt contains all submitted measurements. For FITS, the service emits labeled engineered features and Random Forest probability.

Structured generation
The API asks Gemini for a Boolean assessment and five important features with High/Medium/Low relevance descriptions.

Human-readable result
The interface translates model evidence into plain language so users can connect the classification to transit depth, periodicity, stellar context and data quality.

Intended role
The explanation supports learning and candidate triage. It does not replace follow-up observation, vetting reports or expert confirmation.

Known implementation gap: the manual Gemini route currently maps answers to fixed 85%/15% confidence and defaults missing answers toward “true.” The upload route can return randomized fallback features if the Python service fails. Both behaviors must be removed or visibly labeled before scientific use.

06 Current evidence and reproducibility status

Claim or artifact	Evidence in repository	Status
Interactive system	Working UI routes for manual input, FITS upload, results and 3D exploration.	**Implemented**
FITS feature pipeline	Column discovery, cleaning, median normalization, 2,048-point interpolation and 37-feature extraction.	**Implemented**
Model experiments	Random Forest, 1D CNN, LSTM and MLP code plus serialized checkpoints.	**Implemented**
Reported training scale	README reports 7,638 TESS curves and 95.4% positive labels.	**Needs provenance audit**
Reported performance	README reports CNN validation loss 0.0597; saved RF workflow reports AUC metadata.	**Not independently reproduced**
Complete system accuracy	No candidate-grouped held-out test, external set or calibration analysis is documented.	**Not established**

07 Validation plan and next technical milestones

1 · Dataset audit
Record TIC identifiers, sectors, label source and download date. Resolve Unknown labels and quantify duplicates.

2 · Leakage-safe split
Group observations by target star before splitting so repeat sectors from one object cannot enter both train and test.

3 · Baselines and ablations
Compare RF, CNN, LSTM and MLP against class-frequency and simple depth/period rules. Remove feature families one at a time.

4 · Calibration
Report reliability curves, Brier score and threshold tradeoffs. Replace fixed confidence with measured probabilities.

5 · External validation
Test on a later TESS sector or a separately curated candidate set, preserving untouched labels until final evaluation.

6 · Explanation study
Measure whether students can identify the decisive features, uncertainty and need for follow-up more accurately after using ExoPlain.

Required reporting: sample counts by class · target-grouped train/validation/test split · ROC-AUC and PR-AUC · precision, recall and F1 · confusion matrix · calibration error and Brier score · inference latency · failure-mode examples.

FAILURE MODES THAT THE EVALUATION MUST ISOLATE

Eclipsing binaries
Deep V-shaped or alternating events can mimic planetary transits.

Stellar variability
Spots, pulsations and flares change brightness without an orbiting planet.

Instrument artifacts
Momentum dumps, gaps and background changes can create false structure.

Low-SNR transits
Shallow events test sensitivity, calibration and threshold stability.

Distribution shift
Different sectors, cadences and stellar populations may alter model behavior.

Missing metadata
Incomplete stellar context can weaken both classification and explanation.

Decision rule for the final system: label outputs as candidate-screening evidence, publish calibrated probability with uncertainty, expose whether the result came from the classifier or an unavailable-service fallback, and require human review before any confirmation claim.

TAKE-HOME MESSAGE

ExoPlain connects raw exoplanet evidence, machine-learning predictions and readable explanations in one learning workflow. The next priority is rigorous validation, calibrated uncertainty and removal of simulated fallbacks.

Project sources: ExoPlain Next.js application and API routes; ExoplanetWebsiteFiles/web_fits_feature_server.py; random_forest_exoplanet_model.py; cnn1d_exoplanet_model.py; lstm_exoplanet_model.py; train_python_model.py; project README. **Scientific context:** NASA Exoplanet Archive; TESS mission light curves; TFOPWG candidate dispositions. **Disclosure:** architecture descriptions and reported model values come from the project repository and have not yet been independently reproduced. Project page: spaceappschallenge.org/2025/find-a-team/planet-pioneers2/